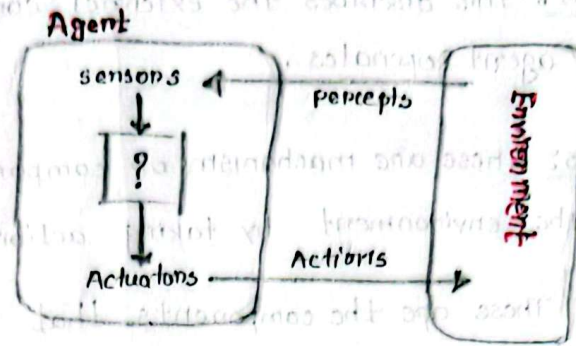


INTELLIGENT AGENT

Agent: An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.



Agent program: The set of programs or algorithms governing the agent's behaviour and decision-making.

Rationality: The ability of an agent to choose actions that maximize the likelihood of achieving its goals. e.g. A smart thermostat....adjusting temperature settings to save energy while maintaining comfort.

Autonomy: The degree to which an agent can operate independently without direct human intervention.

Example: Drone surveying agricultural fields without constant human control.

PEAS

The rationality of an agent, is specified by performance, Environment, Actuators and Sensors. They are grouped together and called PEAS. PEAS specify the task environment. Each letter in PEAS, represent a different aspect of task environment. It stands for

Performance measure: This defines how the success of the agent in the task environment will be assessed.

Environment: This describes the external context on the task environment in which the agent operates.

Actuators: These are mechanisms or components through which the agents affects the environment by taking actions.

Sensors: These are the components that allow the agent to perceive or receive information about the environment.

Agent type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, Keyboard

Fig: PEAS description of the task environment for an automated taxi

Deterministic: If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is deterministic.

Example: A simple calculator performing arithmetic operations.

Stochastic: If the next state of environment is not determined by current state, then it is called.

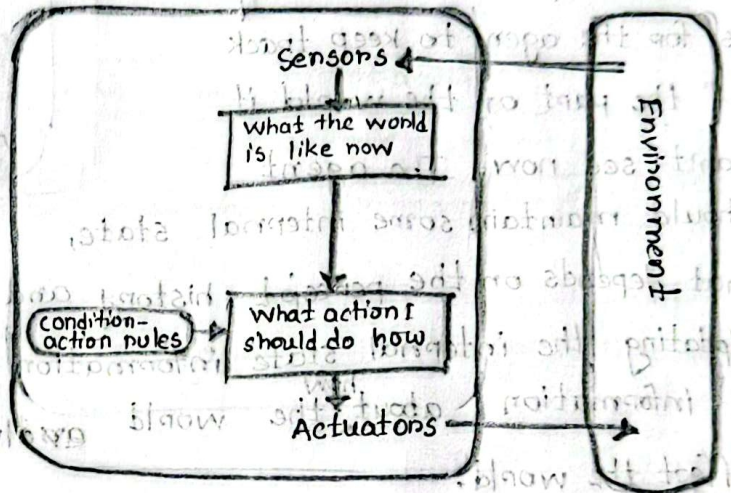
Example: Weather forecast

Agent = Architecture + Program

- Simple reflex agents
- Model-based reflex "
- Goal-based agents &
- Utility-based "

Simple reflex agent: These agents select actions based on the current percept, ignoring the rest of the percept history.

Example: Vacuum agent



function SIMPLE-REFLEX-AGENT(percept) returns an action

persistent: rules, a set of condition-action rules

state $\leftarrow$ INTERPRET-INPUT(percept)

rule $\leftarrow$ RULE-MATCH(state, rules)

Action $\leftarrow$ rule.Action

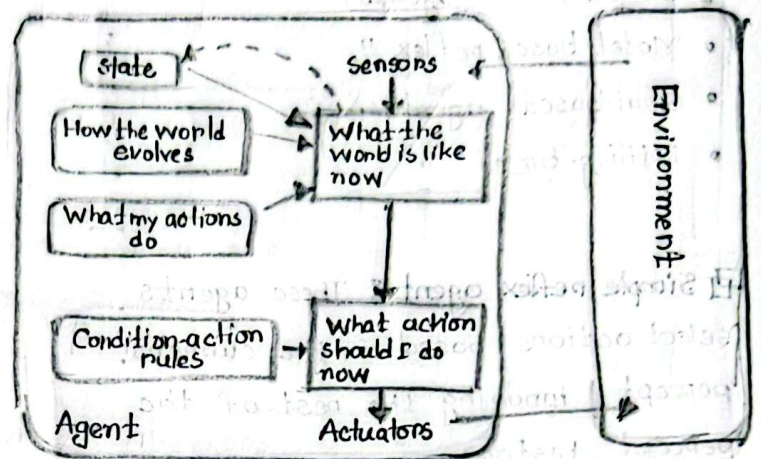
return Action

Algo: It acts according to a rule whose condition matches the current state, as defined by the percept.

Model-based reflex agent

A model based agent is a type of intelligent that uses a model of the world to make decisions and take actions in response to stimuli from its environment. The effective way to handle partial observability is for the agent to keep track of the part of the world it can't see now. The agent should maintain some internal state,

that depends on the percept history and reflects the current state. Updating the internal state information requires two kinds of knowledge, one is information about ^{how} the world evolve and how the agents own action affect the world.



function Model-Based-Reflex-Agent (percept) returns an action

```

state ← INTERPRET-INPUT(percept)
rule ← RULE-MATCH(state, rules)
action ← rule.ACTION
return action
    
```

```

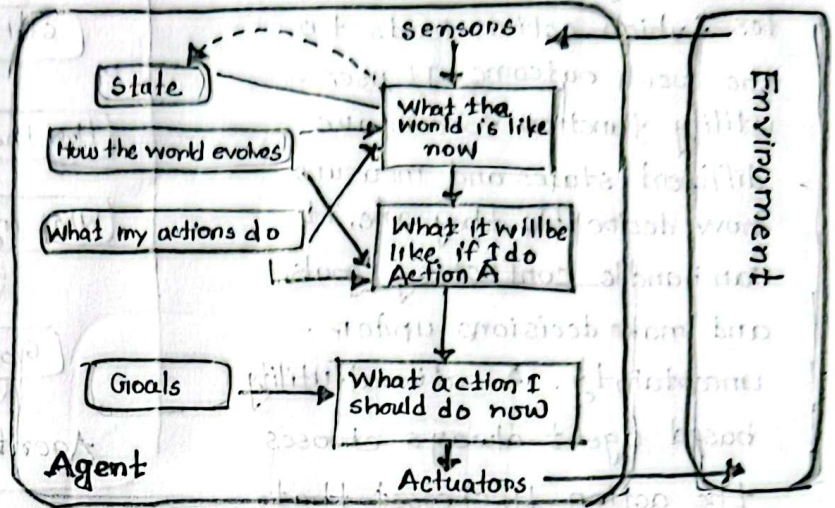
state ← UPDATE-STATE (state, action, percept, model)
rule ← RULE-MATCH (state, rules)
action ← rule.ACTION
return action
    
```


Goal-based Agent

A goal based Agent chooses actions not only from the current state by also considering its goals. It uses a model of the world plus goals to decide which action will achieve the desired outcome. Unlike reflex agents, it can reason about the

future and plan sequences of action.

Goal-based agents are more flexible, because changing the goal automatically changes their behaviour.



function GOAL-BASED-AGENT(percept) returns an action

persistent: state, agent's current conception of the world
model, a description how the world evolves
goal, desirable states

action, the most recent action

state ← Update-STATE(state, action, percept, model)

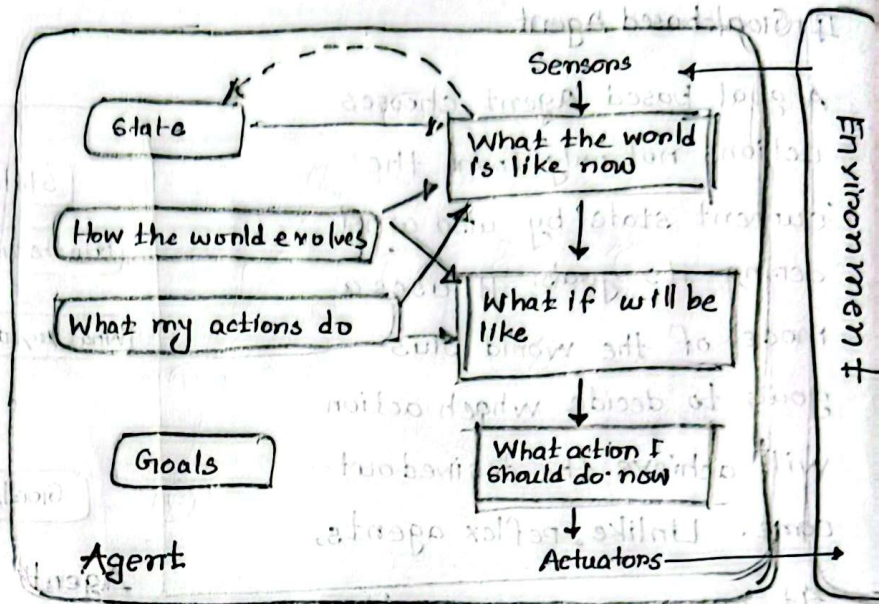
goal ← CHOOSE-GOAL(state, goals)

action ← SEARCH(state, goal, model)

return action

Utility Based

A utility-based agent not only achieves goals but also evaluates which action leads to the best outcome. It uses a utility function to compare different states and measure how desirable they are. It can handle conflicting goals, and make decisions under uncertainty. A rational-utility based agent always chooses the action that maximizes its expected utility.

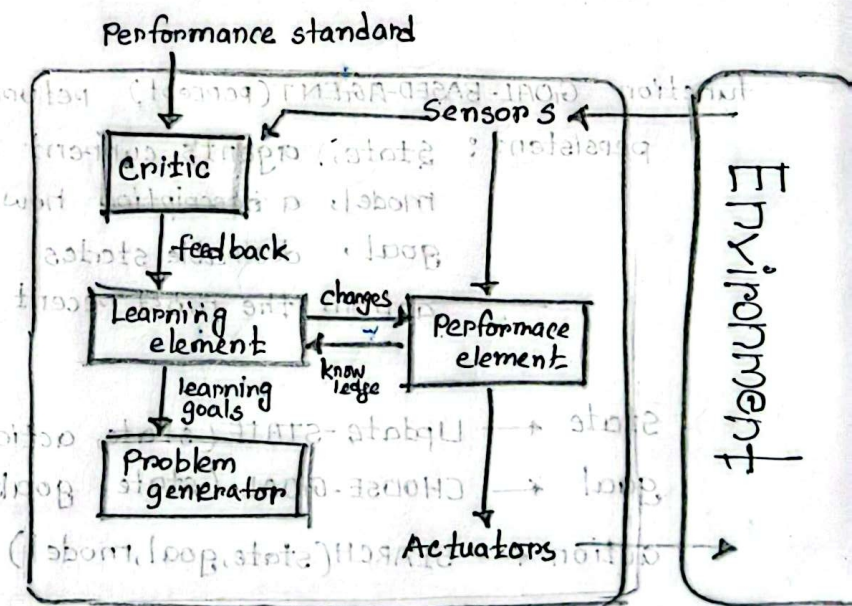


A Learning Agent

- improves its performance over time by gaining experience from the environment.

4 elements

- i. Performance element (chooses action)
- ii. Learning " (improves behavior)
- iii. Critic " (gives feedback)
- iv. Problem Generator (suggest new experience)



Agent program

can represent the environment in three ways

- i. Atomic Representation : Each state is a single invisible unit (black box) e.g: C
- ii. Factored " : A state is described by a set of attributes with specific values (variables + attributes) eg: City, GPS, gas level.
- iii. Structured " : States include objects, their properties & relationships. (Objects + Relationships)
[Car, truck]